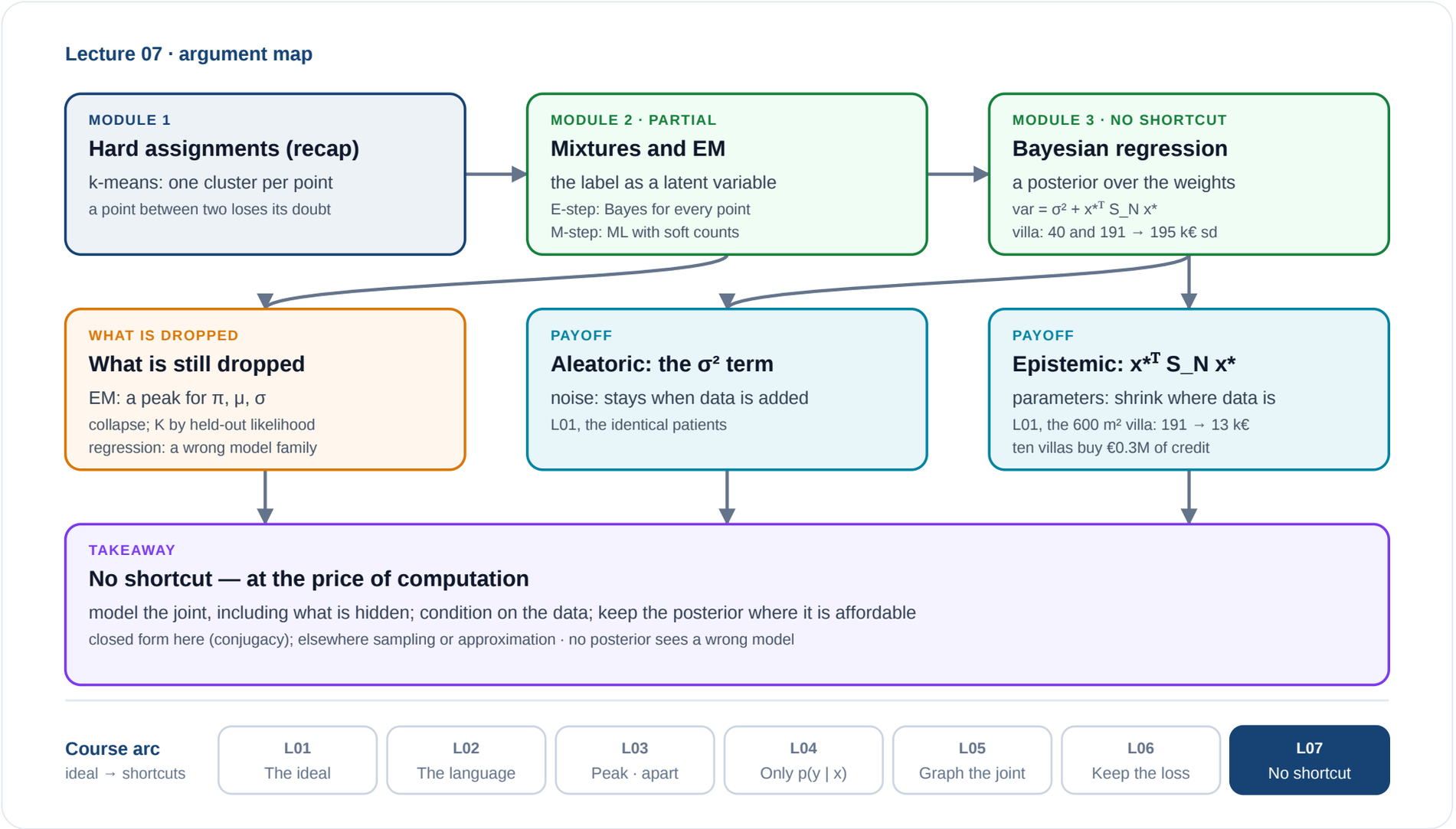
Probabilistic Machine Learning

Lecture 07: Selected Probabilistic Algorithms

Prof. Dr. Michael Granitzer

Argument Map



Module 1

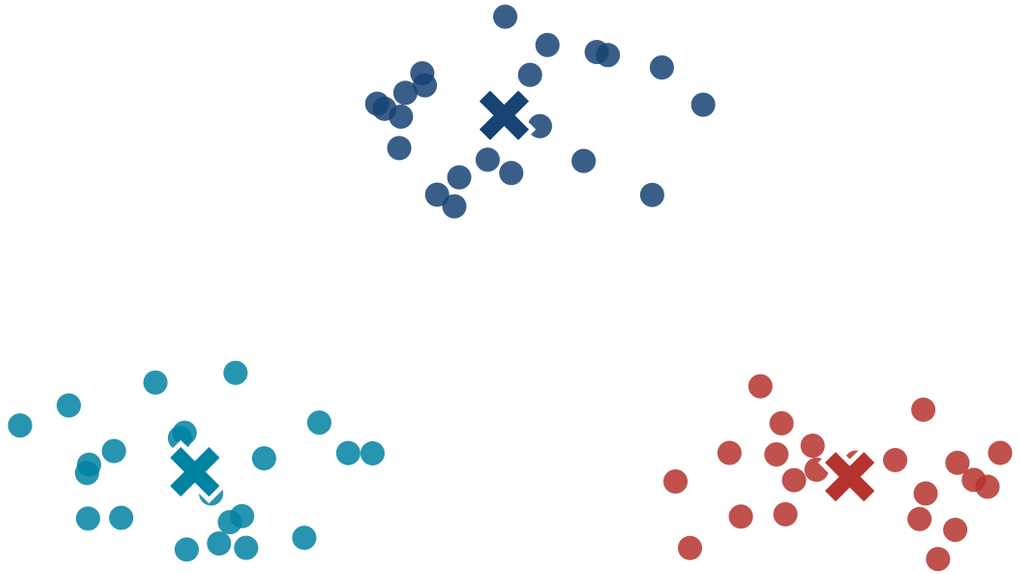
Hard Assignments (Recap)

Guiding Question: k-means gives every point exactly one cluster. What is lost for a point that sits between two?

k-Means in One Slide

Minimise $\mathrm{SSE} = \sum_j \sum_{\mathbf{x} \in C_j} \|\mathbf{x} - \boldsymbol{\mu}_j\|^2$ by alternating: **assign** each point to the nearest centroid, **move** each centroid to the mean of its points. SSE never increases — it stops in a local optimum that depends on the start.

Move each centroid to the mean of its points (round 3)



Where k-Means Fails

failure	what k-means assumes	fixed by a Gaussian mixture?
clusters of different spread	every cluster equally wide (distance only)	✓ each component has its own variance
elongated clusters	round clusters	✓ each component has its own covariance
clusters of different size	equal shares: the boundary lies halfway between centroids	✓ mixing weights π_k
a point between two clusters	one label per point	✓ a probability per cluster
rings, spirals	one blob per cluster	✗ still blobs — needs another representation
how many clusters	k is given	✗ still chosen — but by held-out likelihood

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What **should** an algorithm say?

> k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What **should** an algorithm say?

"About half and half" — which needs the cluster label to be a **random variable** with a distribution, not a value.

> k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

Module 2

Mixture Models and EM

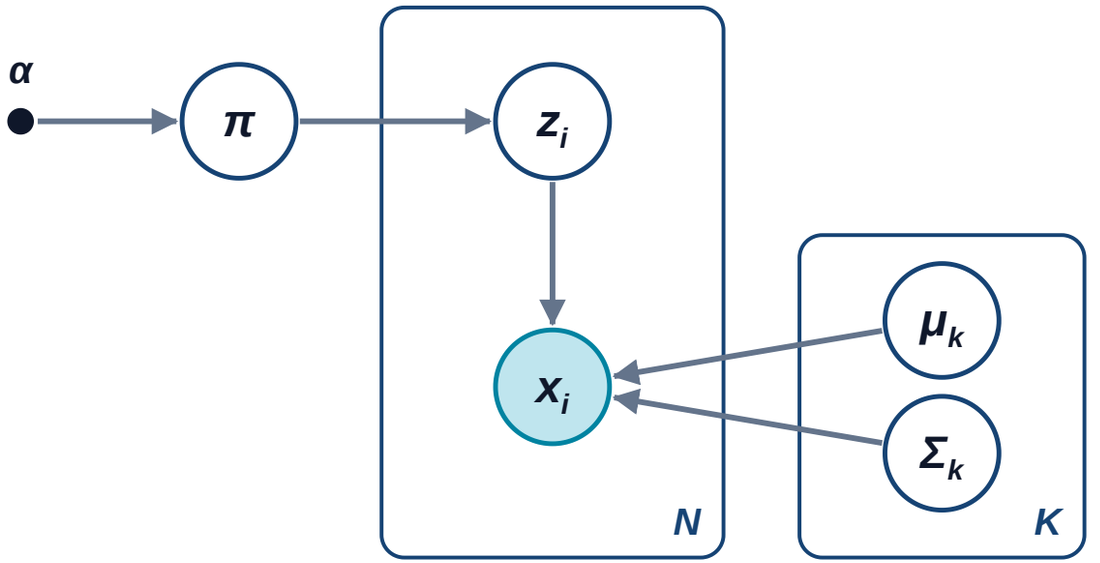
Guiding Question: If the cluster label were just another unobserved variable, could we infer it the way we infer anything else — and what would still be a guess?

The Label as a Hidden Variable

Thirty flats, described by their **price per m²**.
The listing does not say which district a flat is in — suburbs or old town.
A **Gaussian mixture** tells how such data arise:

1. pick a component $z \in \{1, \dots, K\}$ with probability π_z,
2. draw $x \sim \mathcal{N}(\mu_z, \sigma_z^2)$.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$$



The plate of Lecture 05: z_i is **latent** — never observed.

Why There Is No Closed Form

For one Gaussian, maximum likelihood is the sample mean and variance (Lecture 03). For a mixture, the log-likelihood is a **log of a sum**:

$$\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$$

The sum inside the log ties all components together; setting the gradient to zero gives no formula.

If you **knew** each flat's district, how would you fit the two Gaussians? If you **knew** the two Gaussians, how would you find each flat's district?

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If you **knew** each flat's district, how would you fit the two Gaussians? If you **knew** the two Gaussians, how would you find each flat's district?

Known labels: count, average and take the spread per district — maximum likelihood per group. Known Gaussians: Bayes' theorem per flat. **EM alternates the two.**

The E-Step: Bayes for Every Point

Given the current parameters, the **responsibility** of component k for point i is the posterior of the hidden label:

$$\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$$

Prior (the share of component k) times likelihood (how well k explains x_i), normalised over the components.

> The same computation as Lecture 02's disease test — once per data point. A flat halfway between the two districts gets about $0.5/0.5$.

The M-Step: Maximum Likelihood With Soft Counts

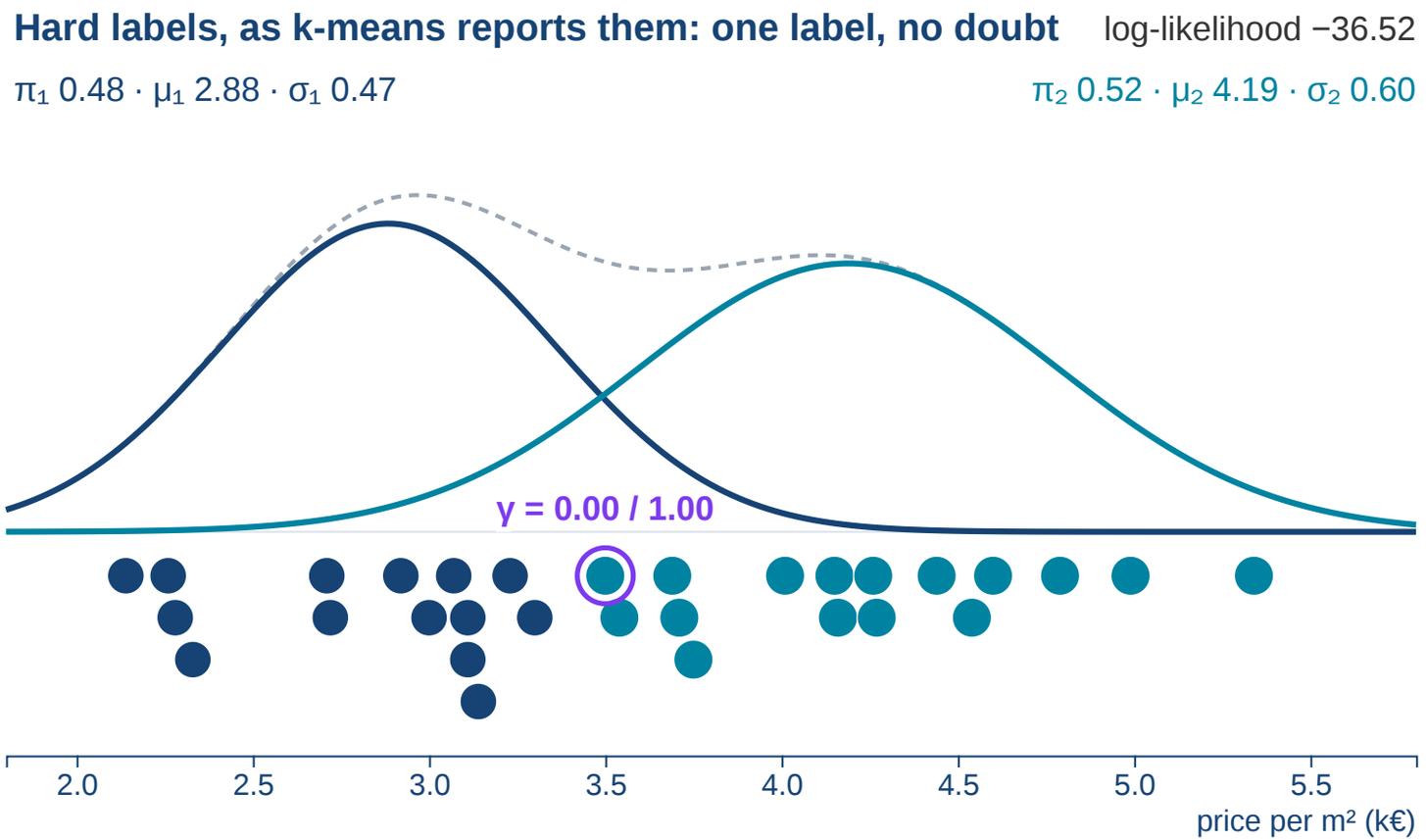
$$N_k = \sum_i \gamma_{ik}, \qquad \pi_k = \frac{N_k}{N}$$

$$\mu_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i, \qquad \sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{ik} (x_i - \mu_k)^2$$

Expectation–Maximisation

1. Start with a guess for π_k, μ_k, σ_k.
2. **E-step:** compute every γ_{ik} (Bayes).
3. **M-step:** refit π_k, μ_k, σ_k — each point counts γ_{ik} times.
4. Repeat until $\ell(\theta)$ stops rising — it never decreases.

Watching EM



Predict before the first click: which flats will come out half-coloured — and does the flat at 3.5 k€/m² end up decided?

EM and k-Means

	k-means	EM on a Gaussian mixture
assignment	hard: one cluster per point	soft: responsibilities γ_{ik}
cluster shape	equal and round (distance only)	own variance (covariance) and weight
objective	SSE	log-likelihood $\ell(\theta)$
output	labels and centroids	a density $p(x)$ and a posterior per point

k-means is the limit of EM with equal, spherical components whose variance shrinks to zero: the responsibilities collapse to 0 or 1, and the M-step becomes "move to the mean".

What EM Still Drops

✓ **The hidden labels:** a full posterior per point — the ideal, carried out.

✗ **The parameters:** π_k, μ_k, σ_k are **one point estimate** — Lecture 03's peak. The plate draws them as unknowns; EM gives them no error bar.

✗ **A peak that may not exist:** a component that shrinks onto one point has $\sigma_k \to 0$ and infinite likelihood. In practice: a floor on σ_k, or a prior (MAP).

✗ **The number of components:** the training likelihood always rises with K. Choose K by the **held-out** log-likelihood (Lecture 06), not by an elbow.

> EM is a **partial shortcut**: Bayes for what is hidden, a peak for the parameters. Keeping a posterior over π, μ, σ as well needs sampling or variational inference.

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

$0.75 \cdot 0.2 = 0.15$ and $0.25 \cdot 0.6 = 0.15$, so $\gamma = 0.5/0.5$. The prior favours component 1 three to one, the likelihood favours component 2 three to one — they cancel. Nearest-centroid assignment would ignore the weights.

Takeaway

A mixture model makes the cluster label a **latent variable with a posterior**. EM alternates **Bayes for the labels** (E) and **weighted maximum likelihood for the parameters** (M): the posterior over what is hidden, a peak for the rest. k-means is what remains when the doubt is squeezed out.

Module 3

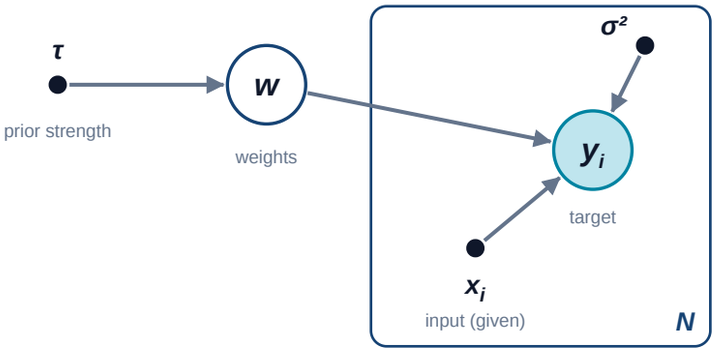
Keeping the Uncertainty

Guiding Question: Your model prices a 600 m² villa at €2.1M — five times larger than any flat it was trained on. Should it say how sure it is, and how much of its doubt could data remove?

Keeping the Distribution

Lecture 01's house-price model: **20 flats of 40–120 m²**, and a question about a **600 m² villa**. The model and the plate of Lecture 05:

$$y_i = w_0 + w_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$



- → **Likelihood:** the Gaussian noise model of Lecture 04, $\sigma = 40$ k€.
- → **Prior:** $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ — intercept within about ±1,000 k€, slope within about ±20 k€/m². Weak.
- → **Posterior:** Gaussian again — the prior is **conjugate**.

Precision Adds Up

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}, \qquad \mathbf{m}_N = \frac{1}{\sigma^2}\mathbf{S}_N\mathbf{X}^\top\mathbf{y}$$

The update is adding

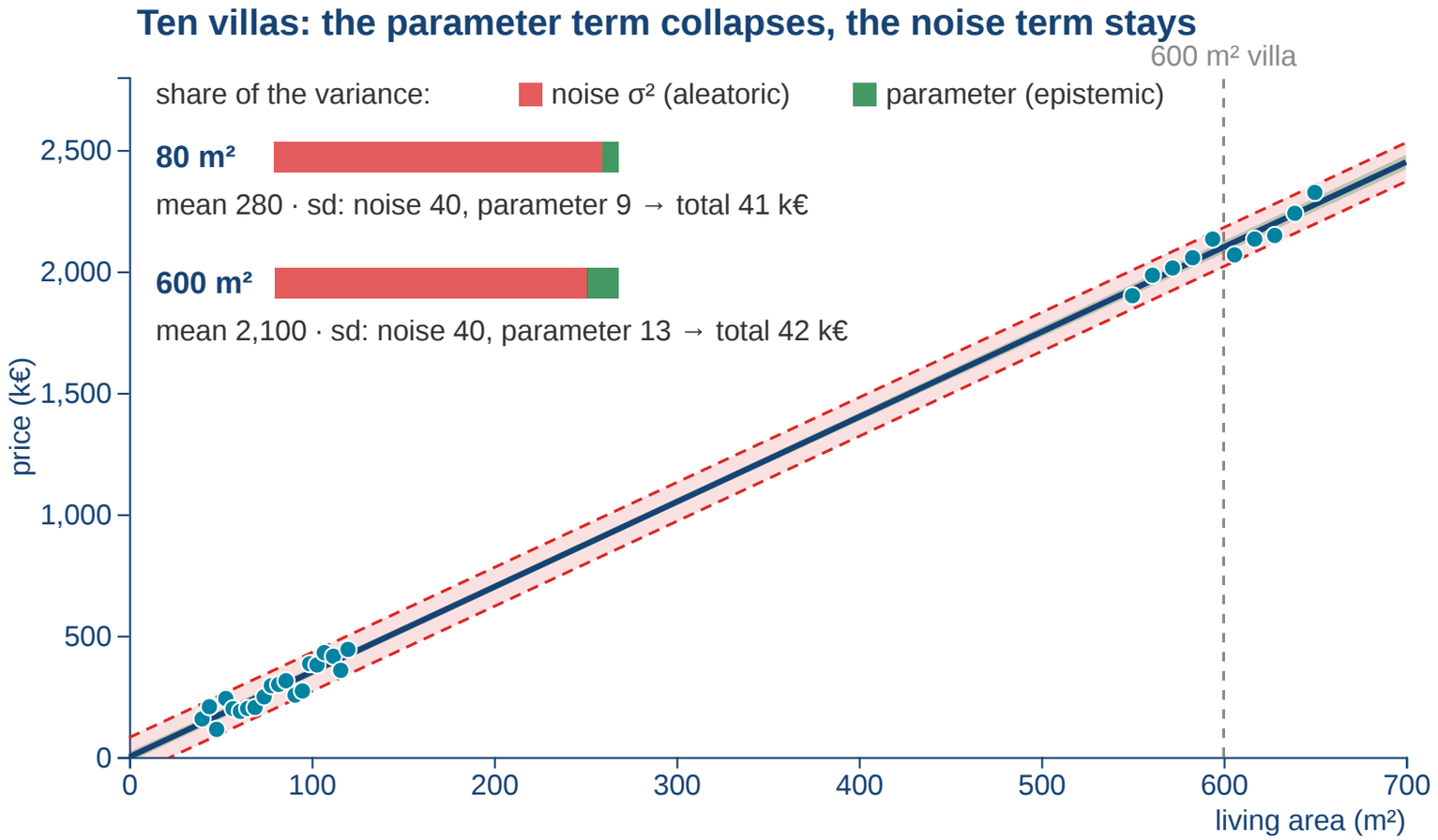
Posterior precision = prior precision + data precision — Lecture 02's **adding counts**, for Gaussians. Every flat adds $\frac{1}{\sigma^2}\mathbf{x}_i\mathbf{x}_i^\top$.

The centre is the ridge estimate

$\mathbf{m}_N$ is the MAP estimate — ridge regression. With this weak prior it is almost least squares: slope **3.496** vs **3.501** k€/m².

What is new is not $\mathbf{m}_N$ but $\mathbf{S}_N$: the slope is 3.50 ± 0.37 k€/m² (one sd) — and that uncertainty is multiplied by 600 m² at the villa.

Watching the Band Form



Predict before the last click: ten villas of 550–650 m² join the data. Which part of the band shrinks — and which part cannot?

Two Terms of One Prediction

$$p(y^* \mid x^*, D) = \mathcal{N}\left(\mathbf{m}_N^\top \mathbf{x}^*,\ \ \sigma^2 + \mathbf{x}^{*\top} \mathbf{S}_N \mathbf{x}^*\right)$$

■ noise term — aleatoric, fixed · ■ parameter term — epistemic, shrinks with data

query	mean	noise sd	parameter sd	total sd
80 m² (inside the data)	280 k€	40	9	**41**
300 m²	1,049 k€	40	81	**90**
600 m² — the villa	2,098 k€	40	191	**195**
600 m², after ten villas	2,100 k€	40	13	**42**

> Near the data the **noise** dominates, at the villa the **ignorance**. Ten villas remove the ignorance; the noise stays — Lecture 01's two examples.

What the Posterior Cannot Know

The band is uncertainty **within the linear model**. Suppose villas actually sell at 5,000 €/m²: the villa is worth **€3.0M** — far outside 2,098 ± 2 · 195 k€, i.e. €1.71M to €2.49M.

The world is noisy

shows up in σ^2, the noise term — accept it, or measure new attributes

Too few data here

shows up in $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, the parameter term — collect data there

The family is wrong

shows up **nowhere** in the formula — check on held-out data where you predict; compare model families

A wide band is an honest "I don't know". A narrow band is not a proof — it can be confidently wrong if the straight line itself is wrong.

Deciding With the Predictive Distribution

A lender lends up to the value it is 97.5% sure the property exceeds: the mean minus two standard deviations.

€198k

an 80 m² flat: $280 - 2 \cdot 41$

€1.71M

the villa today: $2{,}098 - 2 \cdot 195$

€2.02M

after ten villa sales: $2{,}100 - 2 \cdot 42$

Ten comparable sales are worth about **€0.3M of credit**: buying data pays where the **parameter** term dominates. Near the data it buys nothing — the €40k noise stays.

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

B reports about **±390 k€** ($2 \cdot 195$): its parameter term is large this far from the data. Show **B** — not because its centre is better (both are ≈ €2.1M) but because A claims the precision of an 80 m² flat for a house it has never seen anything like.

Takeaway

Bayesian regression keeps a **posterior over the weights** and predicts with a **predictive distribution** whose variance has two terms: the **noise** term (aleatoric) that no data removes, and the **parameter** term (epistemic) that grows off the data and shrinks where data is collected. What neither term covers is a wrong model family.

Module 4

Closing the Arc

Guiding Question: Seven lectures, one ideal, several shortcuts. Which questions can each method you know still answer — and what would it cost to answer the rest?

The Shortcuts, Revisited

Lecture 01's table, with what the course has learned about each row:

shortcut	what it dropped	bought back by	at the cost of
Keep the peak · 03–04	how sure we are about the parameters	the posterior (Module 3)	a prior, an integral
Assume it apart · 03	the dependences assumed away	a graph that draws them	more parameters
Only $p(y \mid x)$ · 04	every question about x	a generative model	modelling x
Graph the joint · 05	the edges not drawn	more edges	costlier inference
Keep the loss · 06	every probability	calibration, log-loss, error bars	held-out data
EM, partial · 07	how sure we are about π, μ, σ	priors, sampling or VI	computation
No shortcut · 07	nothing — within the model family	—	a prior; the family itself

> Every shortcut is a trade of an answer for computation or data. Knowing which one a method takes tells you which question it can no longer answer.

What Comes After

- → **Gaussian processes** — Bayesian regression with infinitely many basis functions: the two-term band of Module 3, without the straight line.
- → **Bayesian neural networks and deep ensembles** — the parameter term for models with millions of weights, approximately.
- → **Sampling (MCMC) and variational inference** — conditioning on the data when there is no closed form; EM is the special case that keeps a posterior only for the hidden labels.

Model the joint, condition on the data, read off answers — and when you take a shortcut, **name it and know what it drops**. A prediction worth acting on says how sure it is, and why: noise, ignorance, or a model that may be wrong.